



Best Model for FMA Audio Genre Classification

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Introduction

There are many machine learning models to be used for classification of audio samples through Mel-Frequency Cepstral Coefficients (MFCC), but which model performs best is relative to the dataset as well as scoring metrics. With the Free Music Archive (FMA) dataset, several models are tested, compared, and analyzed to find the best performing model for genre classification from an audio sample.

Methodology

1. Data Acquisition:

- FMA provides 917 GiB of audio tracks and datasets
- Split into subsets for smaller working dataset
- MFCC's transform raw audio into integers representing human sound

2. Preprocessing:

- Standardization, large range of values to range similar to all values. Mean of 0, standard deviation of 1
- PCA Dimensionality reduction. FMA dataset has multi-index sets, causing a large amount of dimensions. 95% variance
- Label encoding required for XGBoost

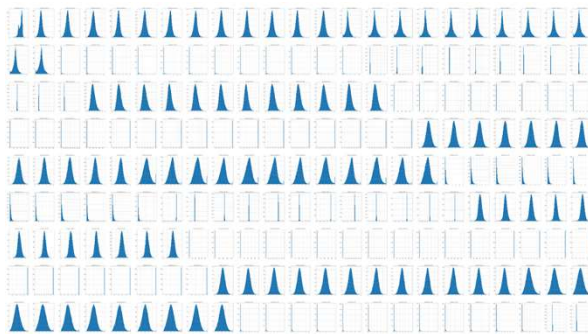


Figure 1. Broad View of Distribution of Data

3. Models:

- SVM – Linear SVC. Better than standard SVC with large datasets
- Random Forest – good with noisy features, many features, independent tree training
- XGBoost – similar to random forest, sequential tree training
- KNN – audio features have clusters similar to what is found
- Neural Network – Multi-layer Perceptron classifier

4. Hyperparameter Tuning:

- 3-Fold Cross-Validated Grid Search

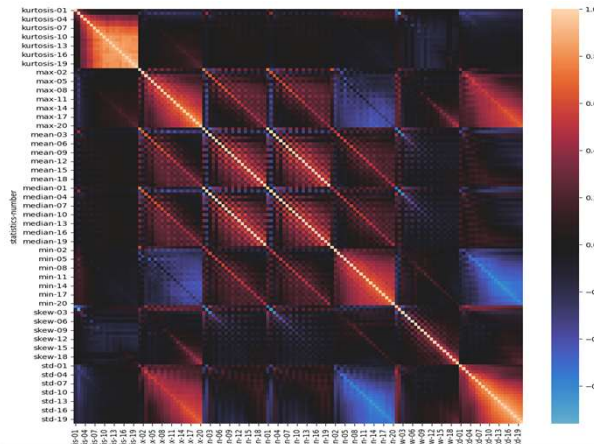
	Blues	Classical	Country	Easy Listening	Electronic	Experimental	Folk	Hip-Hop	Instrumental	International	Jazz	Old-Time / Historic	Pop	Rock	Soul-RnB	Spoken	none
Blues	0	0	0	0	0	0	0	1	0	0	0	0	0	4	0	0	8
Classical	0	31	0	0	1	8	1	0	1	1	0	2	0	0	0	0	42
Country	0	0	1	0	0	0	1	3	0	1	0	0	0	0	0	0	12
Easy Listening	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5
Electronic	0	4	1	0	216	35	4	38	3	3	2	0	2	25	1	1	504
Experimental	0	12	0	0	43	152	12	24	15	4	1	0	4	90	0	1	727
Folk	0	4	2	0	4	15	21	3	0	7	1	1	3	37	0	0	201
Hip-Hop	0	0	1	0	29	2	1	152	1	1	0	0	1	8	0	0	127
Instrumental	0	11	1	0	6	24	4	0	6	0	1	0	1	24	0	0	231
International	1	1	2	0	5	9	5	5	0	17	0	0	0	5	0	0	78
Jazz	0	1	1	0	0	7	0	0	1	0	2	0	0	3	0	0	32
Old-Time / Historic	0	0	0	0	0	1	0	0	0	0	0	52	1	0	0	0	1
Pop	0	0	0	0	21	5	7	6	1	1	0	0	1	24	0	1	137
Rock	0	3	3	0	35	33	9	16	5	7	2	0	13	782	0	4	552
Soul-RnB	0	0	0	0	9	2	0	0	0	1	0	0	0	9	0	0	22
Spoken	0	0	0	0	1	11	2	2	0	1	0	0	0	3	0	4	7
none	0	112	11	4	552	371	124	208	55	50	24	8	35	627	3	13	4115

Figure 2. Confusion Matrix for KNN

5. Metrics:

- Accuracy – plain but helpful for comparisons
- F1 macro – balances precision and recall, good if data is imbalanced
- Balanced - accuracy – covers more than accuracy, average recall per class
- Confusion Matrix – shows correct/incorrect classifications

Figure 3. MFCC Correlations



Results

- Support Vector Machine model performed the worst with the applied parameters, no change after parameter tuning
- Random Forest model performed slightly better than SVM, decrease in accuracy after tuning
- XGBoost performed the second best, score increase after parameter tuning
- K-Nearest Neighbor performed the best for this dataset
- Neural network showed high scores, but due to max iterations not reliable results

Model	Initial Accuracy	Tuned Accuracy	F1 Macro	Balanced-Accuracy	Average
SVM	59.32%	59.32%	12.39%	13.13%	28.28%
Random Forest	59.80%	59.71%	13.07%	12.37%	28.38%
XGBoost	60.46%	60.81%	19.52%	17.47%	32.60%
KNN	49.29%	58.45%	20.37%	19.95%	32.92%
Neural Network	*49.29%	*60.14%	*20.92%	*20.71	*33.92%

Figure 4. Scores of Model Classification

Important Findings:

- Large decrease in score from accuracy to F1 or Balanced-Accuracy
- Neural Network could not converge optimization so scores are not accurate

Conclusions

- ML models each have their own purpose and proper usage
- KNN models classify the best for FMA
- Future Prospects:
 - More model tests
 - Further analysis/preprocessing of data should be explored for improvement of results
 - Alternate hyperparameter tuning

References

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